Protocol Layering & Network Models (OSI vs TCP/IP)

Coverage: Layering Principles, OSI 7-Layer Model, TCP/IP 5-Layer Stack, PDU Encapsulation & Addressing Hierarchy

Prepared by: Prof. Anju Markose | Assistant Professor, Dept. of Computer Science & Engineering | VJCET

1. Protocol Layering Principles

- Modularity & Abstraction: Divides complex networking tasks into decoupled functional layers.
- Encapsulation: Top-down header addition at sender; Decapsulation: Bottom-up header stripping at receiver.
- PDU Hierarchy: L5 Message -> L4 Segment/Datagram -> L3 Packet/Datagram -> L2 Frame -> L1 Bit stream.

2. OSI 7-Layer vs TCP/IP Stack

- OSI Reference Model: 7. Application, 6. Presentation, 5. Session, 4. Transport, 3. Network, 2. Data Link, 1. Physical.
- TCP/IP Stack: Application (integrates L7, L6, L5), Transport (TCP/UDP), Network (IPv4/IPv6), Data Link (Ethernet/802.11), Physical.

■ KTU University High-Yield Exam Questions

- Q: Compare the OSI 7-layer reference model with the TCP/IP 5-layer internet stack. (8 Marks)
- Q: Explain PDU encapsulation and addressing across all 5 layers with a neat diagram. (7 Marks)